

Day 2 Write-Up: Supervised Models on the Adult Income Dataset

Preprocessing comparison, model metrics, and Day 3 plan

1. Preprocessing Choices

The dataset mixes numeric features (age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week) with categorical features (workclass, education, marital-status, occupation, relationship, race, sex, native-country). Each type needs different treatment before a model can use it, so a scikit-learn ColumnTransformer routes each group through its own pipeline, and both pipelines are fit only on the training data to avoid data leakage.

Numeric pipeline: median imputation → StandardScaler

Median imputation was chosen over mean imputation because columns like capital-gain and capital-loss are heavily skewed — most people report 0, while a small number report very large values. The mean would be pulled toward those extreme values, making it a poor stand-in for a “typical” person; the median is more robust to that skew. Dropping rows with missing numeric values was also considered, but none of the numeric columns in this dataset actually contain missing values, so it wasn't necessary — the imputer step is included mainly as a safety net for future or unseen data.

StandardScaler rescales every numeric column to mean 0 and standard deviation 1. This matters most for Logistic Regression, whose regularization penalizes large coefficients — without scaling, a feature like fnlwgt (values in the hundreds of thousands) would be penalized unfairly compared to a feature like age. Decision Trees don't need scaling, but scaling doesn't hurt them either, so sharing one preprocessor for both models keeps the pipeline simple.

Categorical pipeline: most-frequent imputation → OneHotEncoder

Missing categorical values (found in workclass, occupation, and native-country) were filled with each column's most common category. A constant fill value such as “Missing” was also considered — it's arguably more honest, since it doesn't pretend to know the true value — and is planned as an A/B test for Day 3.

OneHotEncoder converts each category into its own 0/1 column (e.g. sex_Male, sex_Female) because both models need numeric input, and one-hot encoding avoids inventing a false order between categories. An OrdinalEncoder (assigning each category a single integer) was considered and skipped, since none of these categories have a natural ranking — a Decision Tree would otherwise treat “Private < Self-emp < ...” as meaningful when it isn't. `handle_unknown='ignore'` ensures that if the test set contains a category never seen in training, it's encoded as all-zeros rather than crashing the pipeline.

2. Model Comparison

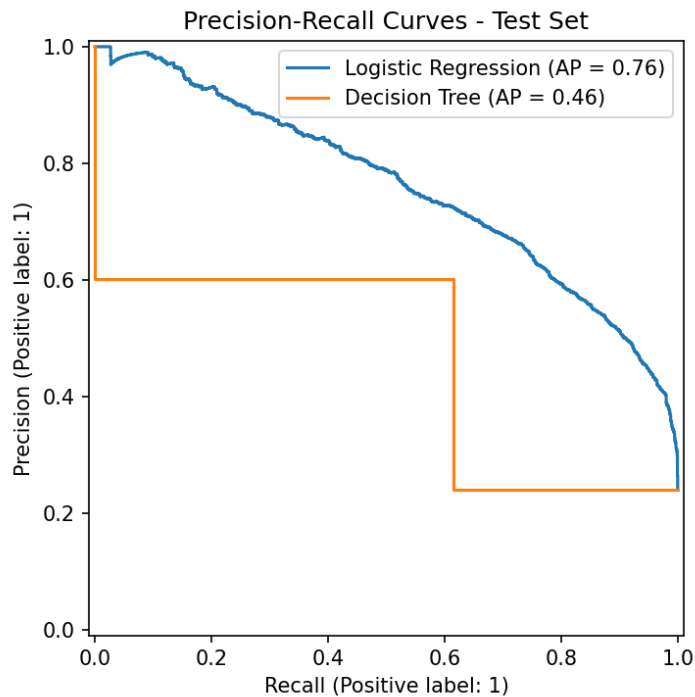
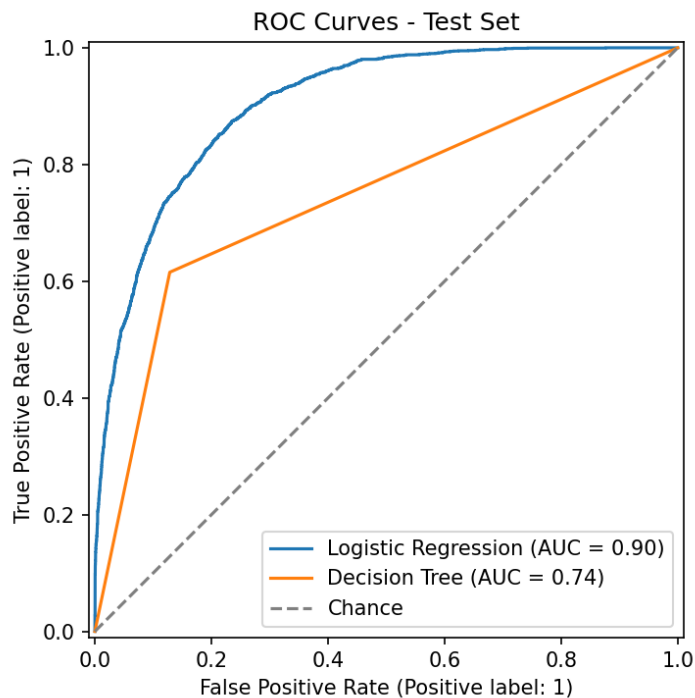
Both models were trained inside a Pipeline (preprocessor + estimator), fit on the training split only, and evaluated once on the held-out test set. The three Day 1 rows below are hand-written rules (not machine-learned, and not re-derived here), included as the floor today's real models need to beat. ROC AUC / PR AUC are marked n/a for them since they output a hard yes/no rather than a probability, so a curve isn't meaningful.

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day 1 - Baseline 1 (majority class)	0.760	0.000	0.000	0.000	n/a	n/a
Day 1 - Baseline 2 (simple rule)	0.742	0.443	0.529	0.482	n/a	n/a
Day 1 - Baseline 3 (advanced rule)	0.724	0.426	0.608	0.501	n/a	n/a
Logistic Regression	0.8507	0.7314	0.5941	0.6557	0.9042	0.7598
Decision Tree (unconstrained)	0.8102	0.6011	0.6155	0.6082	0.7435	0.4620

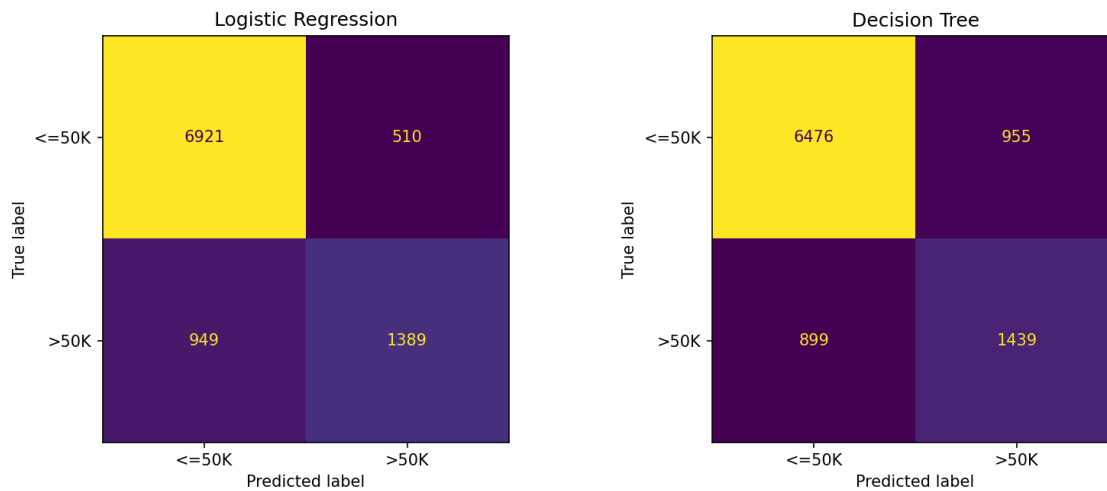
Baseline 3 (the advanced hand-written rule) was the strongest of the three Day 1 baselines, reaching $F1 = 0.501$ by catching 60.8% of true high earners at the cost of a lot of false positives. Both trained models clearly beat all three Day 1 baselines on F1: Logistic Regression reaches 0.656 and the Decision Tree 0.608, a solid jump over hand-written rules without any manual threshold-picking. Between the two learned models, Logistic Regression is the stronger one on every metric except raw

recall, and substantially stronger on ROC AUC (0.904 vs 0.744) and PR AUC (0.760 vs 0.462) — both of which matter more than accuracy here because the classes are imbalanced (about 24% earn >50K).

ROC and Precision–Recall Curves



Confusion Matrices



Logistic Regression: 949 false negatives vs. 510 false positives — it misses a true >50K earner (predicting <=50K) almost twice as often as it wrongly labels a <=50K earner as >50K. Decision Tree (unconstrained): 899 false negatives vs. 955 false positives — the two error types are roughly balanced, with slightly more false positives, a side effect of the tree overfitting to training-set quirks.

Because the dataset is imbalanced toward <=50K, a model can reach a deceptively good accuracy just by leaning toward the majority class — that's exactly what happens with Logistic Regression's lower recall (0.594) than precision (0.731). If this model were used to flag people for a program aimed at higher earners, missing true positives (false negatives) is the costlier silent failure, which is why recall, F1, and PR AUC deserve more weight than accuracy alone when judging these models.

3. Interpretability

Logistic Regression — what drives the prediction

capital-gain (+2.29, the strongest signal) — people with investment income are overwhelmingly in the >50K group. marital-status_Married-civ-spouse (+1.42) and relationship_Wife (+0.92) — married households skew toward higher (often dual) income. occupation_Exec-managerial (+0.80) and education-num (+0.78) — management roles and more years of education both push toward >50K, matching intuition. On the negative side: occupation_Priv-house-serv (-1.59), marital-status_Never-married (-1.12), relationship_Own-child (-1.00), and sex_Female (-0.92) all push toward <=50K.

A handful of small-population countries (France, Cambodia, Italy, Portugal with large positive coefficients; South, Columbia with large negative ones) likely reflect unstable estimates from small sample sizes rather than a strong real signal, and should be treated with caution. More broadly, these are correlations found in 1994 census data, not causal claims — the sex coefficient reflects a real income gap present in the data, not a judgement about any individual.

Decision Tree — depth and overfitting

Left unconstrained (no max_depth), the tree grew to a depth of 59 with 5,703 leaves. Training accuracy reached 0.9999 while test accuracy was only 0.8102 — a gap of about 0.19, a textbook sign of overfitting: with no depth limit, the tree keeps splitting until nearly every training example has its own leaf, memorizing noise instead of general patterns.

The top-level splits are still sensible: the very first split is on marital-status_Married-civ-spouse, followed by capital-gain and education-num — the same features Logistic Regression weighted most heavily. That agreement between two very different model types is a good sign both are picking up genuine structure in the data, not just noise, even though the unpruned tree badly overfits.

4. Model Selection for Day 3

Logistic Regression is the clear choice to continue with for Day 3. The main reasons are:

1. **Strong Performance** - It achieves 0.851 accuracy, 0.731 precision, and a 0.656 F1 score, outperforming all of the hand-written baselines.

2. **Good for Imbalanced Data** - With a ROC AUC of 0.904 and a PR AUC of 0.760, the model performs well despite the class imbalance, where only about 24% of observations belong to the high-income class.
3. **Interpretability** - The coefficients are relatively easy to understand and explain, which is useful if the project needs to communicate how predictions are made to stakeholders.
4. **No Overfitting** - Unlike an unconstrained Decision Tree, Logistic Regression does not show a large training-to-test performance gap, suggesting that it should generalize better to new data.
5. **Simple and Fast** - Logistic Regression is computationally efficient and easier to maintain than more complex models.

Planned Improvements for Day 3

- Try `class_weight='balanced'` to improve recall for the >50K class.
- Test a constant "Missing" category instead of most-frequent categorical imputation.
- Tune the regularization parameter using a simple grid search.

The preprocessing pipeline, consisting of a ColumnTransformer with median-imputed and scaled numeric features and most-frequent-imputed and one-hot-encoded categorical features, has been saved using joblib so it can be reused without rebuilding it.